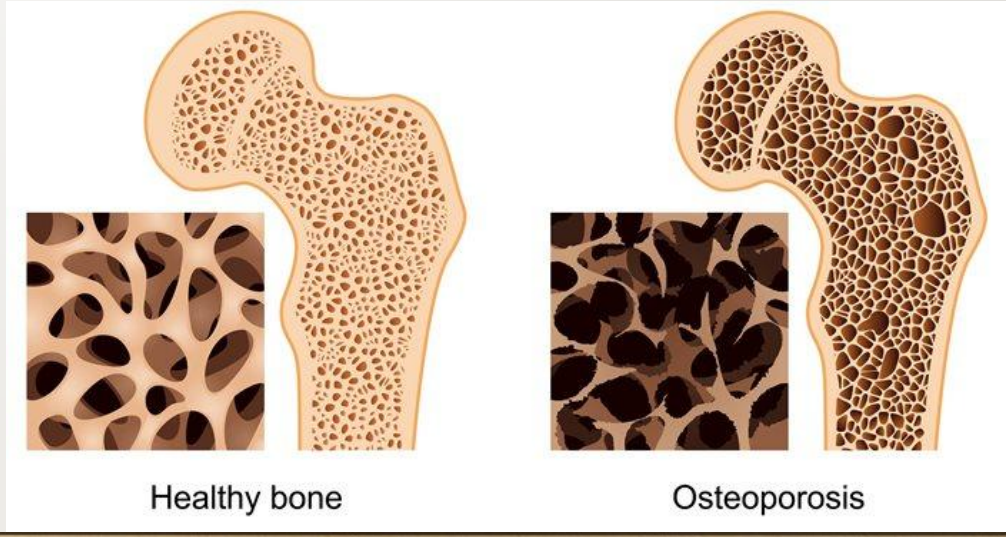
Medication of Osteoporosis

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Objectives

- Osteoporosis and classification (review)
- Drug targets for osteoporosis
- Pharmacotherapy of osteoporosis
  - Antiresorptive drugs:
    - Bisphosphonated
    - Calcitonin
    - Estrogen Replacement Therapy (ERT) /Hormonal Replacement Therapy (HRT)
    - Selective Estrogen Receptor Modulators (SERMs)
  - Bone forming drugs:
    - PTH hormones and analog
    - Calcium/vitamin D
  - Denosumab

Calcium and phosphate homeostasis:

- Three hormones serve as the principal regulators
  - Parathyroid hormone (PTH)
  - Fibroblast growth factor 23 (FGF23)
  - Vitamin D via its active metabolite 1,25-dihydroxyvitamin D (1,25[OH]2D)

Osteoporosis classification

- **Primary osteoporosis**
  - Type I/postmenopausal osteoporosis
  - Type II/senile osteoporosis
- **Secondary osteoporosis**
  - Causes by other conditions as hormonal imbalances, certain diseases or medications.

The causes of osteoporosis

- Genetic
- Age
- Menopause
- Drugs:
- Nutritions status
- No exercise

Drugs that cause bone loss

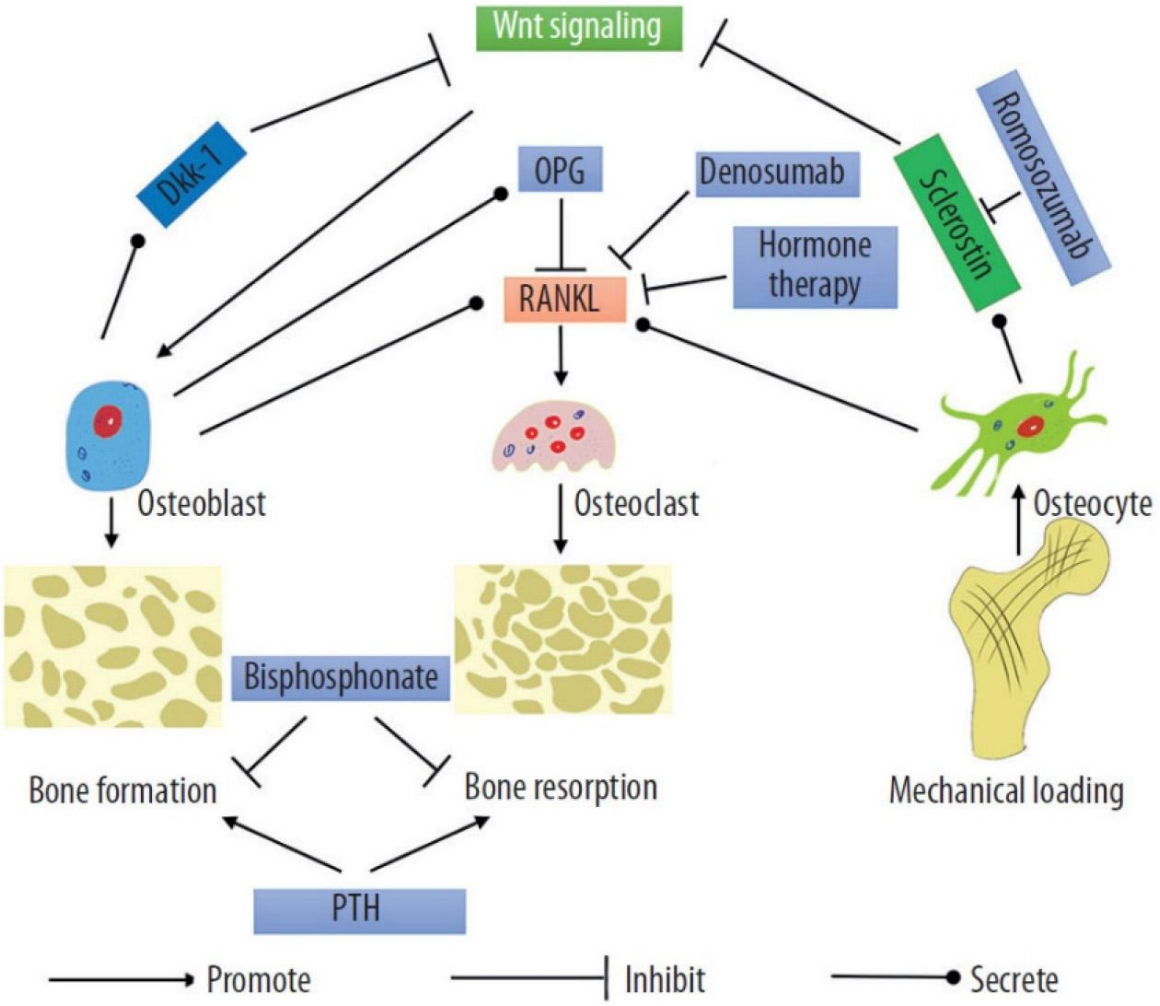
• Aluminum-containing antacids	• Medroxyprogesterone acetate for contraception (Depo-Provera®)
• Antiseizure medicines (only some) such as Dilantin® or Phenobarbital	• Proton pump inhibitors (PPIs) such as Nexium®, Prevacid® and Prilosec®
• Aromatase inhibitors such as Arimidex®, Aromasin® and Femara®	• Selective serotonin reuptake inhibitors (SSRIs) such as Lexapro®, Prozac® and Zoloft®
• Cancer chemotherapeutic drugs: Methotrexate	• Steroids (glucocorticoids) such as cortisone and prednisone
• Cyclosporine A and FK506 (Tacrolimus)	• Tamoxifen® (premenopausal use)
• Gonadotropin releasing hormone (GnRH) such as Lupron® and Zoladex®	• Thiazolidinediones such as Actos® and Avandia®
• Heparin	• Thyroid hormones in excess
• Lithium	

3 main goals of osteoporosis treatments :

1. **Reducing the chance or risk for osteoporotic fracture**.
2. **Reducing or stopping high bone loss** and **promoting construction of bone mass** in order to decrease the possibility for fracture and strengthening of bone.
3. **Improving patient quality of life**.
   - Pain reduction.
   - Maintaining normal or near normal body movement.
   - Increasing self-help ability in daily activity or decreasing dependency.

Drug targets for osteoporosis treatment

- Osteoclast
- Osteoblast
- Parathyroid hormone
- Proteins which involve in osteoclast or osteoblast growth: RANKL, etc.
- Proteins involved *wnt*-signaling pathway→ inhibit osteoblast : sclerostin, dkk-1, etc.



แสดง signaling pathway ในกระบวนการปรับรูปกระดูก (bone remodeling) และยาที่เกี่ยวข้อง (Chang *et al.,* 2018)
ความหมายของคำย่อ: Dkk-1 = Dickkopf-related protein 1, OPG = osteoprotegerin, PTH = parathyroid hormone, RANK = receptor activator of nuclear factor-κB, RANKL= receptor activator of nuclear factor-κB ligand

Preventions

- Good food
- Vitamin and minerals: calcium + vitamin D
- No smoking
- No alcohol
- No caffeine
- Bone density measurement after 50 years old

Pharmacotherapy of osteoporosis

1. **Antiresorptive drugs:**
   - Bisphosphonated
   - Calcitonin
   - Estrogen Replacement Therapy (ERT) /Hormonal Replacement Therapy (HRT)
   - Selective Estrogen Receptor Modulators (SERMs)
2. **Bone anabolic drugs:**
   - PTH hormones and analogs
   - Calcium/vitamin D
3. **Denosumab**

1. Antiresorptive agents: → slow bone loss

I Bisphosphonated : First-line drug

- Alendronate, risedronate, ibandronate (oral)
- Palmidronate, Zoledronate (IV)
- **Mechanism of action:**
  - Inhibit osteoclast function→ ↓ bone resorption and ↓ blood calcium level
  - Inhibit enzyme in mevalonate pathway in osteoclast cell
- **Pharmacokinetics:**
  - Oral drug has low bioavailability → **Take medicine during empty stomach**
  - Mainly excretion via kidney
- **Drug interaction:** calcium and antacid → ↓ **drug absorption**
- **Clinical uses:**
  - **Osteoporosis or osteopenia in postmenopausal women at high risk of fractures (***drug of choice***)**
  - **Paget's disease**
  - **Hypercalcemia of malignancy: IV form**
- **ADRs: bone pain, hypocalcemia, esophagitis, nausea, abdominal pain, etc.**

1. Antiresorptive agents: → slow bone loss

II Calcitonin

- **Mechanism of action:**
  - Bind with G-protein coupled receptor at osteoclast cell surface
  - Inhibit osteoclast function: ↓ bone resorption and ↓ blood calcium level
  - Inhibit calcium and phosphate reabsorption at renal tubule → ↓ blood calcium level
- **Pharmacokinetics:**
  - Route: subcutaneous (SC) or nasal spray
- **Clinical uses: rarely used**
  - Osteoporosis
  - Paget's disease
  - Hypercalcemia
- **ADRs:**
  - Tachyphylaxis
  - Hypersensitivity
  - Nausea
  - Flushing

1. Antiresorptive agents: → slow bone loss

III Estrogen

Mechanism of action:

- Inhibit cytokine gene transcription: IL-6 → ↓ osteoclast growth and differentiation
- Inhibit osteoblast and osteocyte apoptosis

Clinical uses:

- Use for HRT to decrease menopausal symptoms such as hot flush and dry vagina.

- **ADRs:**
  - Breast tenderness
  - Vaginal bleeding
  - Venous thromboembolism
  - Increase risk of endometrial cancer
  - Breast cancer

1. Antiresorptive agents: → slow bone loss

IV Selective estrogen receptor modulators (SERMs)

Mechanism of action:

- Bind with estrogen receptor → SERM-ER complex → bind with hormone response element
- Inhibit osteoclast growth and increase osteoclast apoptosis
- Inhibit osteoblast and osteocyte apoptosis

Clinical uses:

- Raloxifene: estrogen agonist at bone and estrogen antagonist at endometrium and breast tissue.
- Use for osteoporosis without high risk of breast cancer and endometrium cancer

2. Bone anabolic drugs

I Parathyroid or PTH analogs : teriparatide

Mechanism of action:

- Stimulate bone formation by
  - ↑ differentiation of osteoblast
  - ↑ osteoblast function
- ↑ serum calcium

Pharmacokinetics: rapid absorption, Tmax 30 min, half life 1 hr. and duration of action 4 hr.

Clinical uses:

- Osteoporosis in menopausal women
- Fracture risk man
- Low dose via subcutaneous once daily

2. Bone anabolic drugs

II Calcium and vitamin D

- Natural sources: calcium-rich food, sunlight

Ex.

- Milk 250 ml give calcium 300 mg
- A cup of Tofu give calcium 250 mg
- Two spoons of crispy small fish give calcium 226 mg
- Green vegetable 1 ladle give calcium 96 mg

- Vitamin D allows the body to absorb calcium.

Calcium-rich food



Types of calcium for supplement

- Calcium carbonate → calcium 40%
- Tricalcium phosphate → calcium 38%
- Calcium citrate → calcium 21%
- Calcium lactate → calcium 13%
- Calcium gluconate → calcium 9%

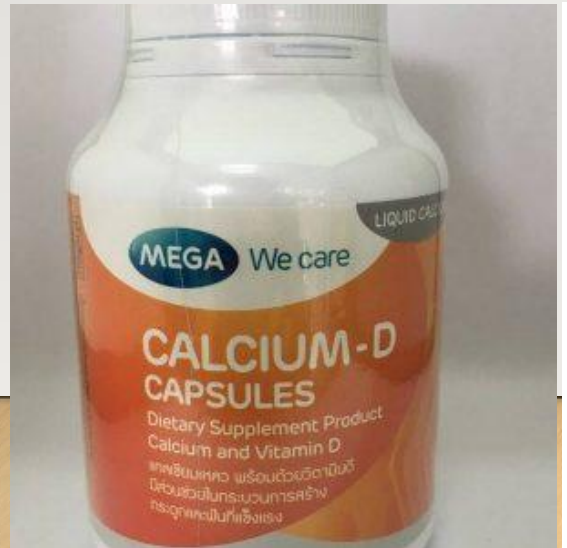
Dosage forms of calcium:

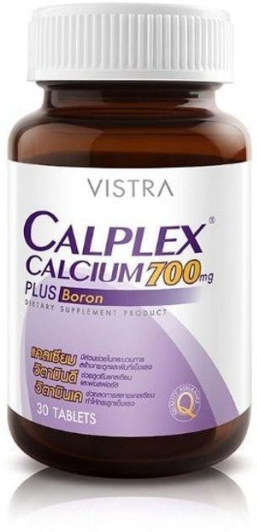
- Tablets
- Effervescent tablets
- Capsules

















Dose and administration

- Calcium carbonate 500 mg/1 tablet give calcium 200 mg, need 2-3 tablets/day
- Calcium citrate 1,000 mg/1 tablet give calcium 200 mg, need 2-3 tablets/day
- Not over than 500 mg of calcium per time
- Aftermeal is the best for absorption with meal
- Avoid taking calcium with tetracycline or ferrous

ADRs:

- Constipation
- Gas in stomach

Vitamin D

- 🌞 **Sources of Vitamin D**
  - **Endogenous synthesis**: UVB radiation from sunlight converts 7-dehydrocholesterol in the skin to **vitamin D3 (cholecalciferol)**.
  - **Dietary intake**: Found in fatty fish, fortified foods, and supplements as **vitamin D2 (ergocalciferol)** or **D3**

Functions of Active Vitamin D (Calcitriol)

- Enhances **intestinal absorption** of calcium and phosphate. ***
- Regulates **bone remodeling** by osteoblasts and osteoclasts. ***
- Modulates **immune function**, **cell growth**, and **inflammation**.
- Influences **neuromuscular** and **glucose metabolism**

Vitamin D metabolic pathway

Liver Hydroxylation

- Vitamin D (D2 or D3) is converted to 25-hydroxyvitamin D [25(OH)D], also called **calcidiol**.
- This is the main circulating form and **the best indicator of vitamin D status**.

Kidney Hydroxylation

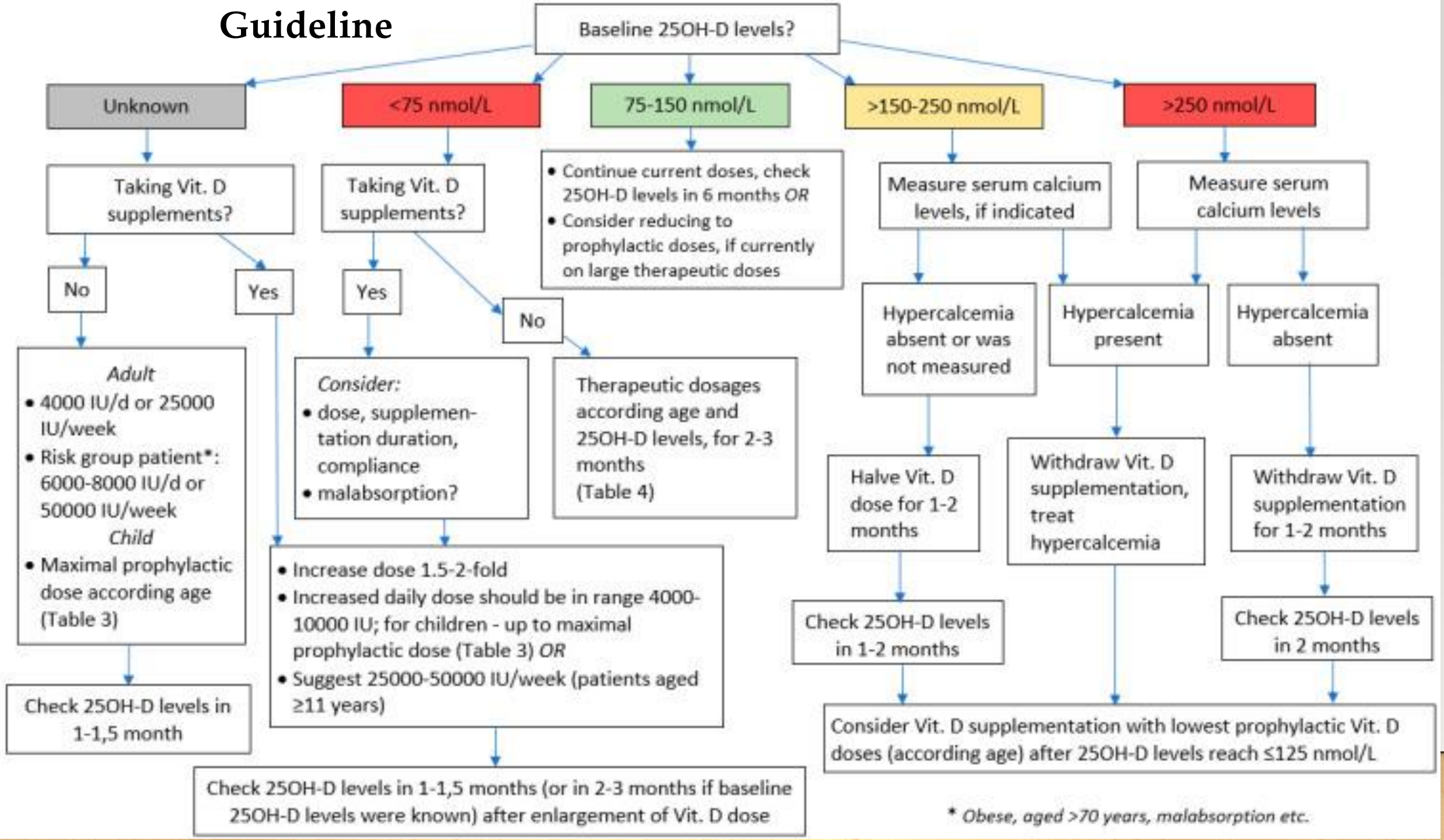
- 25(OH)D is further hydroxylated by 1-α-hydroxylase to form 1,25-dihydroxyvitamin D [1,25$(OH)_2$D], also known as **calcitriol**.
- **Calcitriol is the active form** that binds to vitamin D receptors in various tissues.

Vitamin D assessment level status

Serum 25(OH)D is used to assess vitamin D status:

- <12 ng/mL: Deficiency
- 12–20 ng/mL: Inadequate
- ≥20 ng/mL: Adequate for most people
- 50 ng/mL: Potential for adverse effects

Guideline


Baseline 25OH-D levels?
Unknown
<75 nmol/L
75-150 nmol/L
>150-250 nmol/L
>250 nmol/L
Taking Vit. D supplements?
No
Yes
Adult
• 4000 IU/d or 25000 IU/week
• Risk group patient*: 6000-8000 IU/d or 50000 IU/week
Child
• Maximal prophylactic dose according age (Table 3)
Check 25OH-D levels in 1-1,5 month
Taking Vit. D supplements?
Yes
No
Consider:
• dose, supplementation duration, compliance
• malabsorption?
• Increase dose 1.5-2-fold
• Increased daily dose should be in range 4000-10000 IU; for children - up to maximal prophylactic dose (Table 3) OR
• Suggest 25000-50000 IU/week (patients aged ≥11 years)
Check 25OH-D levels in 1-1,5 months (or in 2-3 months if baseline 25OH-D levels were known) after enlargement of Vit. D dose
• Continue current doses, check 25OH-D levels in 6 months OR
• Consider reducing to prophylactic doses, if currently on large therapeutic doses
Therapeutic dosages according age and 25OH-D levels, for 2-3 months (Table 4)
Measure serum calcium levels, if indicated
Measure serum calcium levels
Hypercalcemia absent or was not measured
Hypercalcemia present
Hypercalcemia absent
Halve Vit. D dose for 1-2 months
Withdraw Vit. D supplementation, treat hypercalcemia
Withdraw Vit. D supplementation for 1-2 months
Check 25OH-D levels in 1-2 months
Check 25OH-D levels in 2 months
Consider Vit. D supplementation with lowest prophylactic Vit. D doses (according age) after 25OH-D levels reach ≤125 nmol/L
* Obese, aged >70 years, malabsorption etc.

Vitamin D (fat soluble vitamin):

Role: Promote calcium absorption

- Vitamin D2 (ergocalciferol) from plant
  - Sources: Mushroom
  - Supplement: dose 25,000-50,000 IU
  - was metabolized to be 25-hydroxyvitamin D2 in liver

Vitamin D

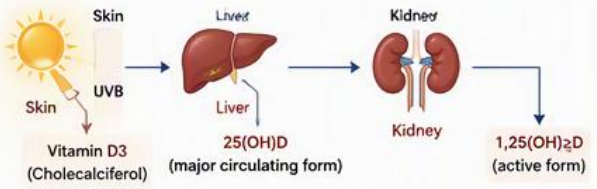
- Vitamin D3 (cholecalciferol) from animal
  - Sources:
    - Sunlight
    - Meal: meat, egg, milk, fatty fish, fish oil, liver, butter
    - Supplements: dose 1,000-10,000 IU
    - was metabolized to be 25-hydroxyvitamin D3 in liver
  - Improve vitamin D (calcifediol) level better than D2

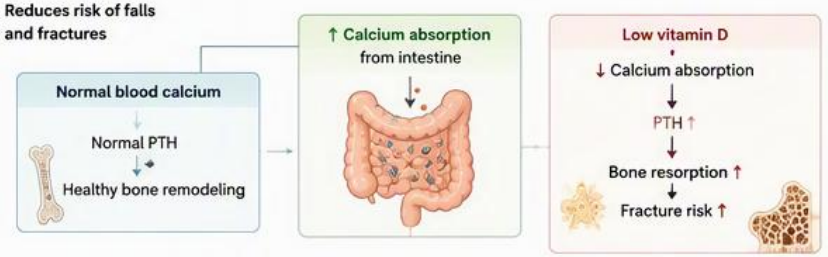
VITAMIN D TREATMENT AND CALCIUM SUPPLEMENT
FOR OSTEOPOROSIS PREVENTION

1. WHY VITAMIN D & CALCIUM ARE IMPORTANT

Role of Vitamin D

- Helps the body absorb calcium from the gut
- Maintains normal blood calcium and phosphate
- Supports muscle function and balance
- Reduces risk of falls and fractures





2. VITAMIN D BLOOD TEST

Test: Serum 25-hydroxyvitamin D [25(OH)D]

- Best indicator of vitamin D status

25(OH)D Level	Interpretation
< 20 ng/mL (<50 nmol/L)	Deficiency
20 – 29 ng/mL (50 – 74 nmol/L)	Insufficiency
≥ 30 ng/mL (≥75 nmol/L)	Sufficient for bone health
> 100 ng/mL (>250 nmol/L)	Risk of toxicity

Target level for osteoporosis prevention:
≥ 30 ng/mL (≥ 75 nmol/L)

3. VITAMIN D TREATMENT

A. Vitamin D Deficiency (< 20 ng/mL)

Loading (repletion) dose
Choose one option

- Vitamin D3 (Cholecalciferol) **50,000 IU** once weekly for 6–8 weeks
- Vitamin D3 **6,000 IU** daily for 6–8 weeks

→ Recheck level in 8–12 weeks

B. Maintenance dose
(After level is ≥ 30 ng/mL)

Vitamin D3
1,000 – 2,000 IU daily
(Older adults or those with risk factors may need 2,000 – 4,000 IU daily)

Special situations

- Obesity, malabsorption, CKD, advanced age → may need higher dose
- Always use Vitamin D3 (cholecalciferol) for better efficacy
- Take with a meal containing fat for better absorption
- Avoid excessive intake (>4,000 IU/day long term unless supervised)

4. CALCIUM FOR BONE HEALTH

Daily Calcium Requirement
(total from food + supplements)

Age	Calcium (mg/day)
19 – 50 years	1,000 mg
> 50 years	1,000 – 1,200 mg
Pregnant / Lactating	1,000 – 1,300 mg

How to take calcium supplements

- Calcium carbonate: take with food
- Calcium citrate: can take with or without food
- Split dose: ≤ 500–600 mg calcium per time for best absorption
- Do not exceed 2,000–2,500 mg/day from all sources

Calcium-rich foods (examples)





Diet should be the primary source. Use supplements to meet the required amount.

5. MONITORING AND SAFETY

Baseline (before or at start)

- Serum 25(OH)D
- Serum calcium
- Phosphate
- Creatinine (kidney function)
- PTH (if indicated)

Follow-up

- Recheck 25(OH)D in 8–12 weeks after starting treatment
- Then every 6–12 months
- Recheck calcium if using high doses or if symptoms occur

Symptoms of Vitamin D Toxicity
(usually with levels > 100 ng/mL)

- Nausea, vomiting
- Poor appetite
- Constipation
- Weakness
- Frequent urination
- High calcium in blood (hypercalcemia)

Avoid toxicity

- Do not exceed 4,000 IU/day long term unless supervised
- Do not combine multiple high-dose products
- Monitor levels and calcium

6. WHY CORRECT VITAMIN D BEFORE OSTEOPOROSIS THERAPY?

- Prevents hypocalcemia especially with bisphosphonates, denosumab, or zoledronic acid
- Improves response to osteoporosis medications
- Ensures optimal bone mineralization

Ensure:

- ✓ 25(OH)D ≥ 30 ng/mL
- ✓ Adequate calcium intake (1,000–1,200 mg/day)

→ Then start or continue osteoporosis medication for best results

7. LIFESTYLE FOR OPTIMAL VITAMIN D AND BONE HEALTH



Safe sun exposure

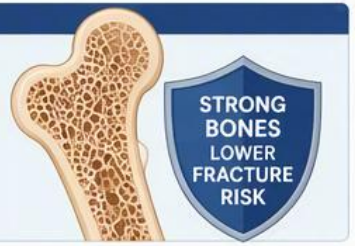
Expose arms and legs to sunlight
10–30 minutes (3–5 times/week)
11 AM – 3 PM
(Depends on skin type, climate, season)

Exercise

- Weight-bearing exercise (walking, hiking, dancing)
- Strength training 2–3 times/week

Other healthy habits

- Do not smoke
- Limit alcohol intake
- Maintain healthy body weight
- Fall prevention (safe home, balance exercises)

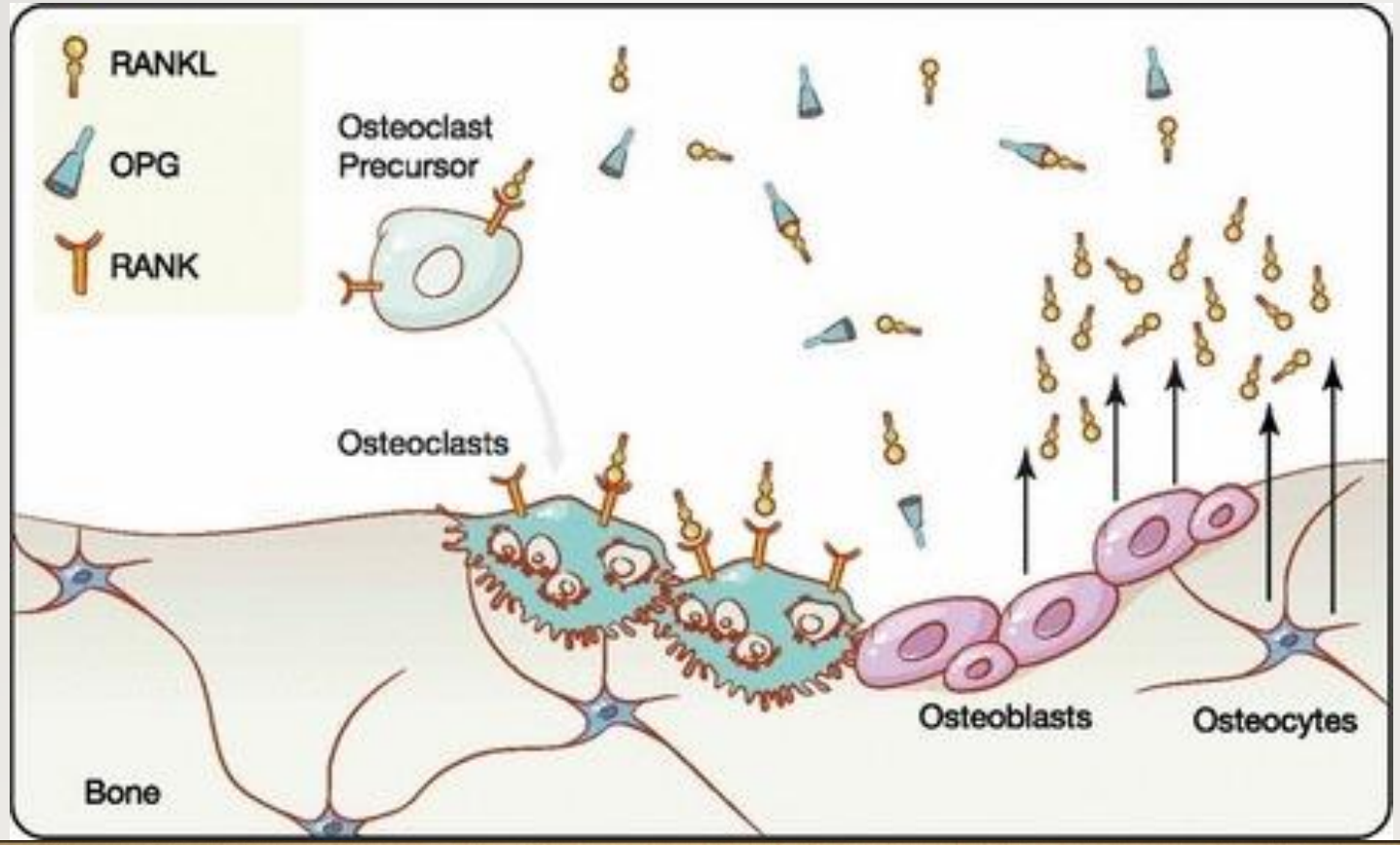


KEY TAKEAWAY: Adequate vitamin D (≥ 30 ng/mL) and calcium (1,000–1,200 mg/day) are essential foundations for osteoporosis prevention.
Combine proper supplementation with healthy lifestyle, exercise, and fall prevention for stronger bones and fewer fractures.

3. Denosumab (monoclonal antibody, Prolia®, Xgeva®)

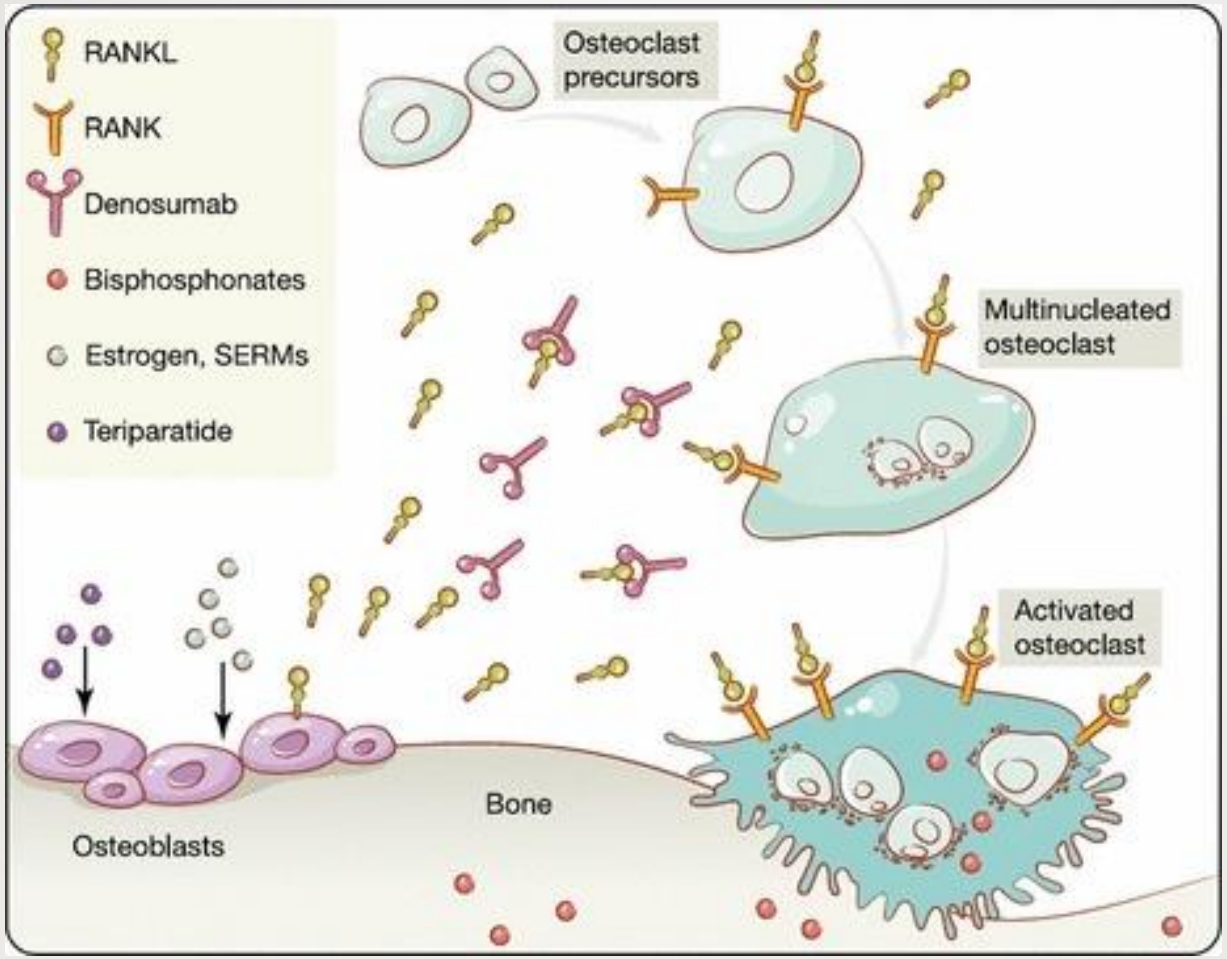
- **Mechanism of action:**
  - Binds with high affinity and specificity to circulating RANKL
  - Inhibit osteoclast formation, function and survival
- **Pharmacokinetics:**
  - Rapid onset
  - Fully reversible effect
  - $T_{1/2}$ = 26 days
  - Clear by reticuloendothelial system

Bone resorption



Reference: D A Hanley, J D Adachi, A Bell, and V Brown. (2012). Denosumab: mechanism of action and clinical outcomes.. Int. J. Clin. Pract. Dec; 66(12): 1139–1146.

Mechanism of action of denosumab



Reference: D A Hanley, J D Adachi, A Bell, and V Brown. (2012). Denosumab: mechanism of action and clinical outcomes.. Int. J. Clin. Pract. Dec; 66(12): 1139–1146.

3. Denosumab (monoclonal antibody, Prolia®, Xgeva®)

- **Clinical uses:**
  - Regimen: Prolia® 60 mg s.c., q 6 months + calcium 1000 mg/day and at least 400 IU vitamin D daily)
  - Postmenopausal osteoporosis
- **ADRs:**
  - Prolia®: back pain, extremity pain, musculoskeleton pain, cystitis, vertigo, etc.
  - Xgeva® : fatigue/asthenia, hypophosphatemia, nausea, dyspnea, diarrhea, hypocalcemia
- **Contraindication:**
  - Prolia®: pregnancy

4. Sclerostin inhibitors

- Humanized IgG2 monoclonal antibody
- Mechanism of action:
  - Inhibit sclerostin → anabolic effect
- Examples: Romosozumab, Blosozumab, etc.
- Indication: Menopausal women who has high risk of fractures

OSTEOPOROSIS
PREVENTION AND THERAPY MANAGEMENT

1. LIFESTYLE & DIET (PREVENTION)



ADEQUATE NUTRITION

- **Calcium** 1000–1200 mg/day (dairy, green vegetables, fish with bones, tofu, nuts)
- **Vitamin D** 800–1000 IU/day (sunlight, fatty fish, eggs, fortified foods)



REGULAR EXERCISE

- **Weight-bearing:** walking, jogging, dancing, stair climbing
- **Muscle strengthening:** resistance training
- **Balance & flexibility:** tai chi, yoga



AVOID RISK FACTORS

- Do not smoke
- Limit alcohol (≤1 drink/day)
- Maintain healthy body weight
- Avoid excessive caffeine



FALL PREVENTION

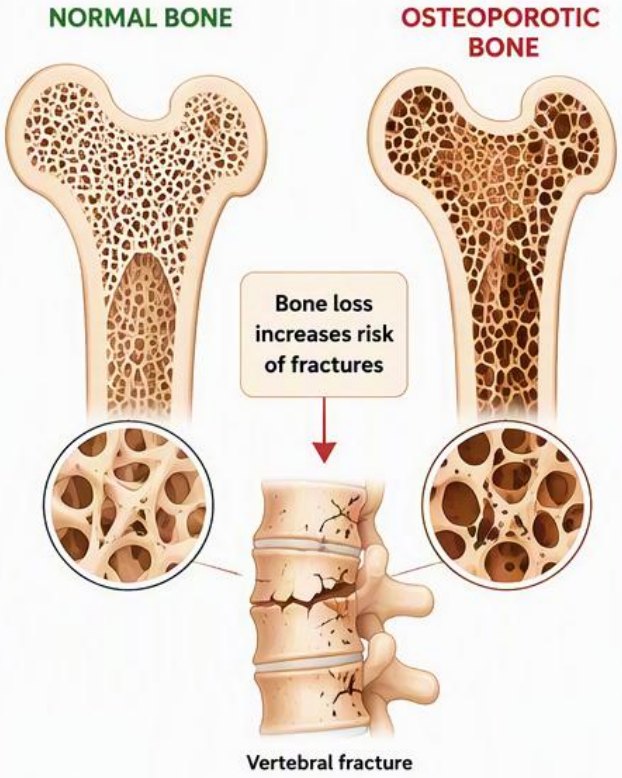
- Adequate lighting, remove hazards
- Use handrails, non-slip mats
- Regular vision & hearing check
- Review medications that cause dizziness

2. SCREENING & EARLY DETECTION



DXA scan

- Bone mineral density (DXA) testing:
  - Women ≥ 65 years
  - Postmenopausal women <65 with risk factors
  - Men ≥ 70 years or with risk factors
- Assess fracture risk (FRAX®)



TREATMENT GOALS

- Prevent fractures
- Increase or maintain bone density
- Improve quality of life
- Minimize side effects

3. THERAPY MANAGEMENT

Assess fracture risk & BMD

Low to Moderate Risk	High/Very High Risk
Lifestyle + Calcium + Vitamin D Monitor BMD	Lifestyle + Calcium + Vitamin D Pharmacologic therapy

ANTIRESORPTIVE THERAPIES (Decrease bone resorption)

Bisphosphonates (First-line)

- Alendronate (oral)
- Risedronate (oral)
- Ibandronate (oral/IV)
- Zoledronic acid (IV)

Reduce vertebral, hip and non-vertebral fractures

Denosumab (RANKL inhibitor)

- Subcutaneous injection every 6 months
- Reduces vertebral, hip and non-vertebral fractures

Selective Estrogen Receptor Modulators (SERMs)

- Raloxifene
- Reduces vertebral fracture risk
- Useful in younger postmenopausal women

Menopausal Hormone Therapy

- Estrogen (± progestin)
- Prevents bone loss
- Consider for menopausal symptoms (individualize risk–benefit)

ANABOLIC THERAPIES (Increase bone formation)

Teriparatide (PTH 1-34)

- Daily subcutaneous injection
- For high risk patients (usually up to 2 years)

Abaloparatide (PTHrP analog)

- Daily subcutaneous injection
- For high risk patients (up to 2 years)

Romosozumab (Sclerostin inhibitor)

- Monthly subcutaneous injection (12 months)
- Increases bone formation and decreases resorption

4. MONITORING & FOLLOW-UP

- Reassess fracture risk periodically
- Ensure adherence and correct use of medications
- Monitor BMD (usually every 1–2 years)
- Monitor calcium, vitamin D and other relevant labs
- Reinforce lifestyle, fall prevention and exercise

Key Point: A combination of adequate nutrition, regular exercise, fall prevention, and appropriate medication (specific to fracture risk) is the most effective strategy to prevent fractures and maintain strong bones.